Investigating Speech Production in Aphasia: A Usage-Based Approach

Jieun Kim[1*], Mara Breen[1], and Rosemary Varley[2]

[1] Department of Psychology, Mount Holyoke College
[2] Department of Language and Cognition, University College London

Background

- Understanding how language is structured and processed in the human mind has been a central quest in various perspectives
  - **Generative Grammar** attributes pauses to a core deficit in syntactic processing, viewing them as a failure to construct complex grammatical structures [4]
  - **Usage-Based Theory** proposes that pauses reflect the cognitive effort of retrieving less frequent and less mentally entrenched linguistic patterns, linking disfluency to retrieval difficulty rather than a damaged grammar module [3]

Bello-Lepe et al. (2024)

This study replicates and extends the work of Bello-Lepe et al. (2024), who investigated speech pauses from a usage-based perspective. They found that for both individuals with aphasia (IWA) and non-brain-damaged (NBD) speakers, pause durations decreased within more strongly collocated word pairs. This effect was significantly stronger in the IWA group. The results provide clear evidence for a usage-based account, suggesting that predictable, frequently used phrases are processed more easily as holistic "chunks," particularly by speakers with aphasia.

Method

Participants

Speech samples were obtained from the AphasiaBank corpus [6] for 34 participants across three groups: individuals with Broca's aphasia, individuals with Conduction aphasia, and non-aphasic controls.

Materials

The corpus analyzed here consists of 31,525 words (8,521 Broca's, 5,616 Conduction, 17,388 Non-Aphasic) with an average log unigram frequency of 14.07 (3.93-17.73). Log bigram frequencies averaged 9.08 (3.04-15.47)

Participant Demographics by Group

Type	N	Age M(SD)	WAB AQ M(SD)
Broca	9	57.9 (15.8)	61.2 (5.2)
Conduction	13	63.3 (14.4)	73.9 (6.9)
Non-aphasic	12	58.7 (12.2)	93.9 (3.1)

Analyses

- **Data Processing**: Speech was transcribed in Praat [2] to measure word and pause durations (pauses > 250 ms). Unigram and bigram frequencies were extracted from the Corpus of Contemporary American English (CoCA) [5] and log-transformed for analysis.
- **Statistical Modeling**: After removing outliers, data were analyzed using Linear Mixed-Effects Models (LMERs) in R. The models included:
  - Outcomes: Pause Duration, Word Duration
  - Predictors: Bigram Frequency, Group, & their interaction
  - Random Effect: Participant (to control for individual variability)

Questions & Hypotheses

Overall Group Comparisons

- **H1 (Lexical Frequency):** Broca's speakers will rely most on high-frequency words
- **H2 (Collocation Strength):** Broca's > Conduction > Non-Aphasic
- **H3 (Pause Duration):** Longest in Broca's, shortest in Non-Aphasic speakers

Effects of Linguistic Predictability

- **H4 (Pause Duration)**: Stronger collocation strength → shorter pauses
  - Effect size: Broca's > Conduction > Non-Aphasic
- **H5 (Word Duration)**: Stronger collocation strength → shorter word durations
  - Effect size: Broca's > Conduction > Non-Aphasic

Results

Overall Group Comparisons

- Lexical Frequency (H1): Conduction speakers produced significantly lower-frequency words than Broca's speakers (p = .024).
  - Non-Aphasic speakers did not significantly differ from Broca's (p = .079).
- Collocation Strength (H2): No significant group differences in bigram frequency (p > .09 for all comparisons).
- Pause Duration (H3): Silent pause duration did not significantly differ across groups (p ≥ .36 for all comparisons).

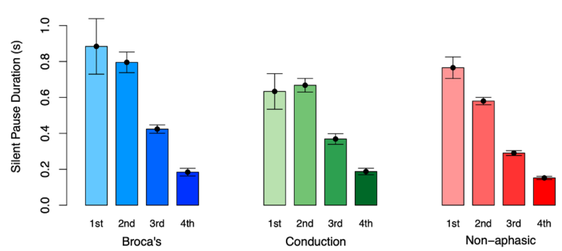

Figure 5. Mean Silent Pause Duration by Bigram Frequency Quartile across Groups. The figure displays mean silent pause duration (in seconds) across four quartiles of Bigram Frequency for each participant group. Error bars represent ±1 standard error of the mean. Pause duration decreases as Bigram Frequency increases, with the steepest decline observed in the Broca's Aphasia group.

Results (cont'd)

Effect of Collocation Strength on Silent Pause Duration

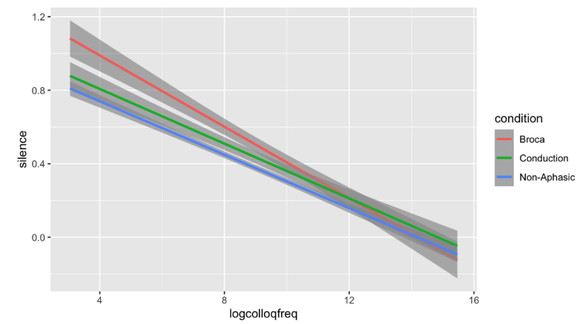

Figure 1. Silent pause duration as a function of log-transformed bigram frequency across Broca's, Conduction, and Non-Aphasic groups. Lines represent model-estimated slopes; shaded regions indicate ±1 standard error.

- Stronger bigram frequency predicted shorter pauses across groups (p < .001).
- The effect varied by condition (interaction p < .05).
- Contrary to prediction, Non-Aphasic speakers showed the strongest sensitivity to collocation strength.

Effect of Collocation Strength on Word Duration

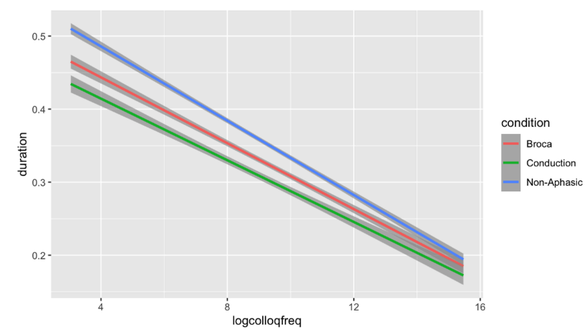

Figure 2. Scaled word duration (z-scored within participant) as a function of log-transformed bigram frequency across the three participant groups. Lines represent model-estimated slopes; shaded regions indicate ±1 standard error.

- Higher bigram frequency predicted shorter word duration across groups (p < .001).
- Broca's speakers showed the steepest reduction in scaled word duration.
- Conduction speakers exhibited a significantly flatter slope (p = .046), indicating reduced articulatory facilitation from predictability.

Discussion

- **Frequency effects:** Higher lexical and bigram frequency predicted shorter pauses and faster word production across all groups, showing sensitivity to predictability.
- **Usage-Based Theory:** Results align with experience-dependent, probabilistic models; speech planning relies on entrenched lexical and multiword units rather than abstract rules alone.
- **Aphasia patterns:** Broca's speakers showed more reliance on high-frequency words, but facilitation was not consistently stronger across all measures, challenging the idea of overreliance in IWA.
- **Language architecture:** Findings suggest a distributed, adaptive language network where frequency effects persist post-stroke, but lesion location, network reorganization, and cognitive constraints shape outcomes.
- **Takeaway:** Speech emerges from the interaction of probabilistic experience and distributed neural architecture, rather than purely rule-based or purely frequency-driven mechanisms.

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